

Authors' Instructions: Preparation of Camera-Ready Contributions to SCITEPRESS Proceedings

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Keywords: The paper must have at least one keyword. The text must be set to 9-point font size and without the use of bold or italic font style. For more than one keyword, please use a comma as a separator. Keywords must be titlecased.

Abstract: The abstract should summarize the contents of the paper and should contain at least 70 and at most 200 words. The text must be set to 9-point font size.

1 Introduction

Artificial Intelligence (AI), and Generative AI (GenAI) in particular, has found a plentiful applications in nearly all aspects of everyday life. On several occasions, the underlying tasks can be formulated as search problems, in the sense of having an AI agent move around a state space, starting from an initial state, crafting a path towards a goal state. Examples of this might include route finding, puzzle solving, shopping planning, code debugging, clinical counseling and so on. Furthermore, restrictions are put on the shape of the *optimal* path, in accordance to the nature of the task at hand, e.g., seeking a shortest path for route finding (Liu, 1996; Migel et al., 2024), finding a particular kind of solution to a puzzle (Mutalova et al., 2023), or strictly following a therapist's guidance to provide further advice to a client (Sedlakova and Trachsel, 2023; Fiske et al., 2019).

Notably, in all such cases, path optimality can also be understood in the context of conformance to the search policy of a coach, be it explicit, e.g., in a therapeutic setting, or implicit, e.g., in the case of route finding. This kind of formulation allows one to also control under appropriate learning semantics, not only *what* the learner learns to accomplish, but also *how* it does so, which might be of higher significance in particular settings. In this work, we put forward such a search coaching paradigm in a simplified set-

ting, exploring this interplay between the end and the means to it learned by a machine learner.

2 Related literature

Knowledge exchange between a coach and a machine learner in the form of context-driven advice giving based on the learner's decisions has been described as soon as in (McCarthy, 1959). Recently, it has been formalized in the context of *Machine Coaching* (Michael, 2017; Michael, 2019) under the Probably Approximately Correct (PAC) model (Valiant, 1984).

A similar approach, eXplainable Interactive Learning (XIL), (Teso and Kersting, 2019) also considers a human supervisor in the role of a learning "coach", with advice either correcting mislabeled data, or wrong explanations provided by the learner for otherwise correct predictions. Nevertheless, while in Machine Coaching coach advice can be directly embedded into the learner's knowledge base in an elaboration-tolerant manner (McCarthy, 1998), XIL utilizes an active learning algorithm as a black-box (Settles, 2009) alongside a local post-hoc explainer (Ribeiro et al., 2016). The corresponding explanations are then used to generate additional labeled data for the learner, to help it cure any misunderstandings.

User feedback can also be utilized to proactively reduce the need for data point labeling, as in (Raghavan and Allan, 2007) for Support Vector Machines. In the same spirit, (Settles, 2011) proposes an interactive framework allowing users to label selected data points

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with the aim of further polishing the training of a document classifier, while in (Zaidan et al., 2007), a machine learning paradigm utilizing user-provided explanations to reduce learning complexity is proposed. Interestingly, in (Zaidan et al., 2007) it is stressed that an alternative to immense data load for performance improvement is *data richness*, i.e., the provision of more meaningful data, e.g., in the form of a correct label accompanied by an explanation for it.

Coactive Learning, proposed in (Shivaswamy and Joachims, 2015), is another interactive Machine Learning framework where a learner receives user advice that is, inherently, only “slightly better” than the learner’s current prediction. The main benefit of the approach, as shown in (Shivaswamy and Joachims, 2015), is that this approach can be integrated into many existing black-box machine learning methodologies.

Given the recent spike in the use of Large Language Models (LLMs) in virtually all domains, human-machine interaction methodologies that utilize human expertise to further polish a LLM have been proposed. In (Welz and Lanquillon, 2024) a human knowledge infusion methodology is presented, where expert users provide domain specific knowledge that is used to populate a knowledge base, then utilized by the system in tandem with the LLM to generate responses to user queries.

3 Sorting coaching

In this Section we empirically explore search coaching, by having a machine learner iteratively seek advice by a coach, as in (Markos et al., 2022), with the aim of iteratively improving its search capacity, guided by the coach’s explicit search policy. Given a fixed goal state, g , the learner applies its search policy, P_{learner} , to identify a path, p , starting from an arbitrary state, s , and ending at a goal state, g . Upon lack of knowledge, the learner seeks advice from its coach, which responds according to its own policy, P_{coach} . Each piece of advice has the form of an “if-then” rule, where the “if” part is state-dependent and the “then” part is always an action to be undertaken by the learner. Rules are prioritized by the coach as in previous works (Markos et al., 2022; Markos and Michael, 2022; Ioannou and Michael, 2021) with the last received ones overriding previous knowledge. A *coaching step* is a single learner request for advice and the corresponding coach response, while a *coaching session* comprises of the full learning cycle, where the learner eventually crafts one such path or the process is aborted (cf Algorithm 1).

Data: $s, g, \text{learner}, \text{coach}$

Result: *learner* with updated search policy.

$\text{advice} \leftarrow \text{null};$

do

$\text{learner.update_hypothesis}(\text{advice});$

$\text{path} \leftarrow \text{learner.search_path}(s, g);$

$\text{advice} \leftarrow \text{coach.evaluate}(\text{path}, s, g);$

while advice is not null;

Algorithm 1: Coaching process. s, g denote the start and goal states respectively.

3.1 Experimental setup

We have implemented two machine coaches, C_{bubble} , which uses bubble sort as its underlying coaching policy, and C_{quick} , which uses quicksort for that purpose, both resulting to advice of the form of “what would the next step in the sorting algorithm be?”. A state of size n in this case is any permutation of $\{1, 2, \dots, n\}$, start states might be arbitrary and the goal state is always $(1, 2, \dots, n)$. Rules in this case have a potentially partial permutation of $\{1, 2, \dots, n\}$ as their condition and a swap action between two items of this permutation, as shown in Figure 1. We let n range from 1 to 20, running $m = 100$ iterations for each value of n . We have also explored two different types of advice for each coach: (A1) *full advice*, where each piece of advice depends on the entire state, and; (A2) *partial advice*, where each piece of advice depends on the specific part of the state to be improved. Moreover, for each advice type we have considered three learner configurations regarding advice memory: (M1) *no memory* across different iterations for the same state size, n , i.e., coaching sessions are pairwise independent; (M2) *short memory* across different iterations for the same state size, n , i.e., coaching sessions across different values of n are independent but not within the same value for n , and; (M3) *long memory* across all values of state size, n , so all coaching sessions are correlated, building on knowledge from previous test cases. In all cases the learner seeks advice upon lack of knowledge for further searching, returning a partial path p starting from s , while the coach adopts one of the following *coaching styles* as illustrated in Figure 2.: (C1) in *reactive* coaching, the coach provides advice on learner’s last wrong decision in p ; (C2) in *proactive* coaching, the coach provides advice on learner’s first wrong decision in p ; (C3) in *reflexive* coaching, the coach provides advice at a random wrong decision in p .



Rbf :: (3,4,1,5,2) implies swap(4,1);
Rbp :: (-,4,1,-,-) implies swap(4,1);
Rqf :: (3,4,1,5,2) implies swap(3,2);
Rqp :: (3,-,-,-,2) implies swap(3,2);

Figure 1: >>> To be completed...

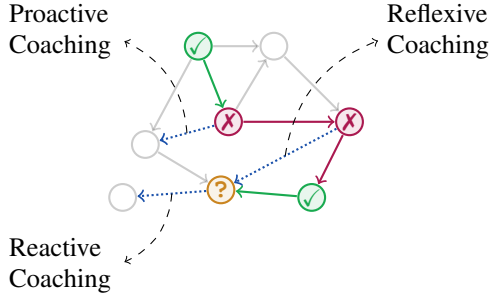


Figure 2: Different coaching styles used in our experimental setup. Upon the learner reaching a state from where there is no knowledge on how to proceed (?), a reactive piece of advice resolves this lack of knowledge, a proactive piece of advice addresses the first wrong decision in the learner’s search path, while a reflexive piece of advice addresses a random mistake / lack of knowledge of the learner.

3.2 Results

In Figures 3 we present the number of coaching interactions required to reach the goal state in each case as a function of the state size, n for all three types of coaching. Considering reactive coaching and the full advice (Fig. 3a), the utilization of either short or long term memory does not appear to have any particular effect on the number of coaching steps, which was expected given the restricted applicability of provided advice. On the contrary, partial advice (Fig. 3d) combined with both short and long term memory result to fewer coaching steps, which can be explained by the broader applicability of provided advice. Also observe that in the case of full advice, coaching steps are proportional to the big- O complexity of the underlying policy / sorting algorithm, which, again, comes as a result of limited advice applicability, leading to the learner receiving step-by-step guidance in each separate case. However, the introduction of both types of memory alongside broader advice allows the learner to generalize beyond the scope of the coach’s policy. Thus, especially in the long memory case (Fig. 3d),

coaching steps for both C_{bubble} and C_{quick} are essentially the same (dotted lines).

Considering proactive coaching, where the learner receives advice on its first wrong search decision (Figs. 3b, 3e), we observe that full advice imposes the same trends in coaching step growth as with reactive coaching. However, providing partial advice, results, again, to the same pattern of coaching steps collapsing under the light of memory, as the learner is able to generalize from previous examples / cases. Furthermore, coaching steps are systematically higher from the corresponding reactive coaching cases (Figs. 3d, 3e). This reflects the more “meticulous” way a proactive coach interacts with the learner, providing (even partial) advice that is more accurate, preventing any unnecessary exploration of the search space, at the cost of having to interact more frequently with the learner, compared to a more “lazy” reactive coach.

Regarding reflexive coaching (Figs. 3c, 3f), we observe similar trends as in proactive coaching, i.e., the coach providing advice addressing earlier learner mistakes on average, guides the latter more accurately even in the case of partial advice. We also report in that case the number of coaching sessions (indicated as *trivial runs*) where at least one advice was provided more than once by the coach, which stems from the randomness in which the learner’s mistake is chosen by the coach.

>>> A few more comments on the differences of the two coaches.

4 Coaching fidelity

Motivated by the high accuracy with which the learner applies the coach’s policy in the case of full advice (Fig. 3, top row), we explore how much the learner conforms to the coach’s instructions in the various settings explored in our case. To that end, we define *coaching fidelity* to be the relative frequency of learner decisions conforming to coach decisions on the same states over all such decisions in a coaching session. We sample $m = 20$ repetitions for state size $n \in \{5, 10, 15, 20\}$ and compute coaching fidelity for each one of those for C_{bubble} , C_{quick} , all coaching styles, and memory configurations.

In Figure 4 we show average coaching fidelity as a function of state size, n . Regardless of memory configuration, full advice and reactive coaching style results to a virtually fully conformant learner, replicating the coach’s policy. Similar results are also obtained for proactive and reactive coaches. However, with partial advice, coaching fidelity depends on the coaching style and the corresponding memory config-

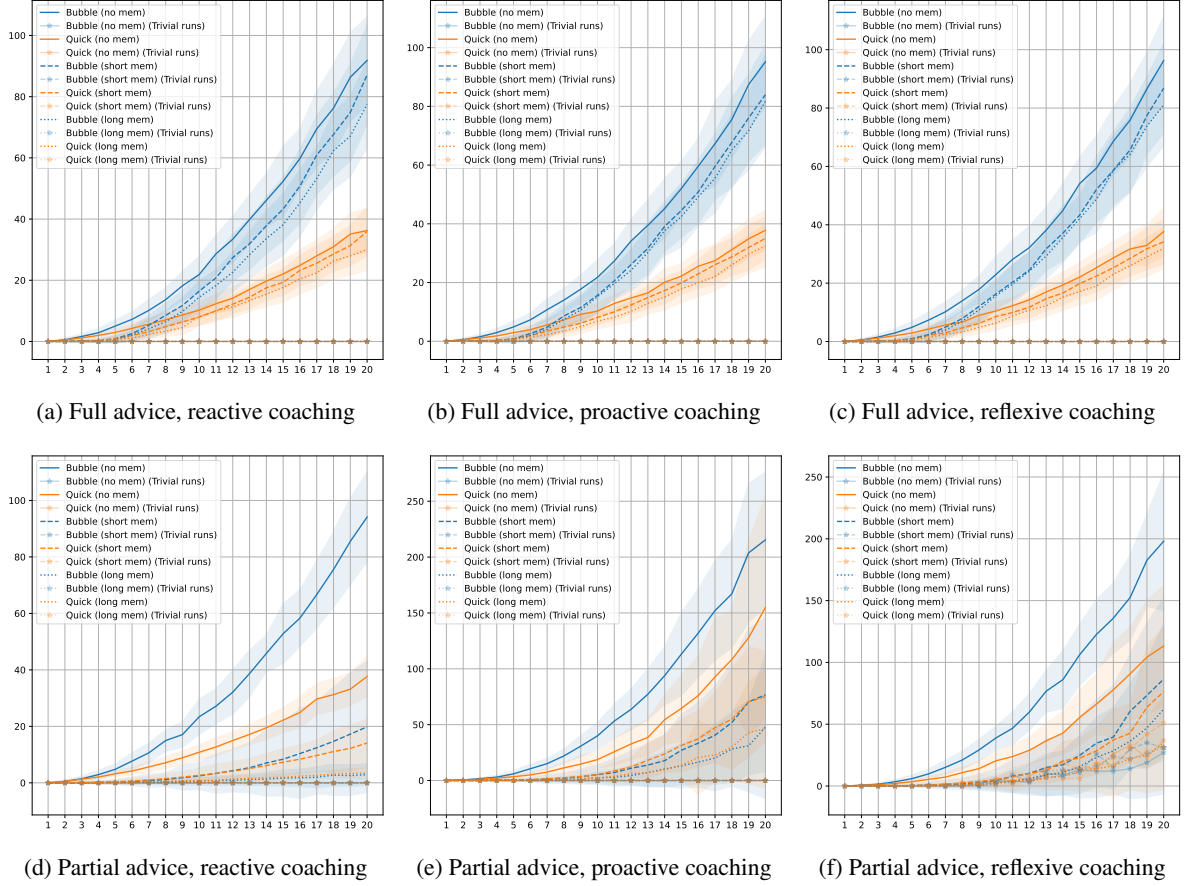


Figure 3: Coaching steps as a function of state size, n . Full and partial advice (top and bottom, respectively) with reactive, proactive and reflexive coaching styles are considered (left, middle, and right, respectively), with all possible coach (C_{bubble} , C_{quick}) and memory (none, short, long) combinations are shown.

uration. In particular, in the case of reactive coaching, partial advice results to virtually vanishing fidelity for all memory configurations. For proactive coaching, however, partial advice results to a virtually constant high fidelity for all examined state sizes, barely affected by the corresponding memory configuration. Similarly, reflexive coaching also results to higher fidelity than reactive coaching, roughly replicating results for proactive coaching. Also, observe how in all cases coach type (C_{bubble} or C_{quick}) does not affect fidelity, except for partial advice reactive coaching, where any differences appear to vanish as state size grows.

Overall, Figure 4 hints towards a trade-off between advice specificity and coaching fidelity. The more specific the advice the more conformant the learner to the coaching policy and, conversely, the more loose the advice, the more the learner explores the state space, utilizing previous knowledge and generalizing. To shed more light onto that interplay, we

introduce a new type of advice, partial@k , $2 \leq k \leq n$, where $k - 2$ randomly selected state variables are included into the advice condition, alongside the two required to describe the swap action of interest. Such advice captures intermediate specificity between the two extremes explored so far, with partial@2 corresponding to partial and partial@n to full advice. We also define *advice coverage* as the percentage state variables that is included in a the condition of a piece of advice, so that, e.g., a piece of advice being triggered for $(x_1, x_2, x_4) = (2, 0, 3)$ for state size $n = 5$ has an advice coverage of $3/5 = 60\%$.

We ran $m = 20$ iterations for state sizes $n \in \{5, 10, 15, 20\}$ and $2 \leq k \leq n$ in each case, for all coaching types, styles, and memory configurations. In Figure 5 we present coaching fidelity against advice coverage for all those cases. Observe, at first, how coach type and memory configuration do not affect coaching fidelity, while the most significant effect comes from different coaching styles. Indeed, in

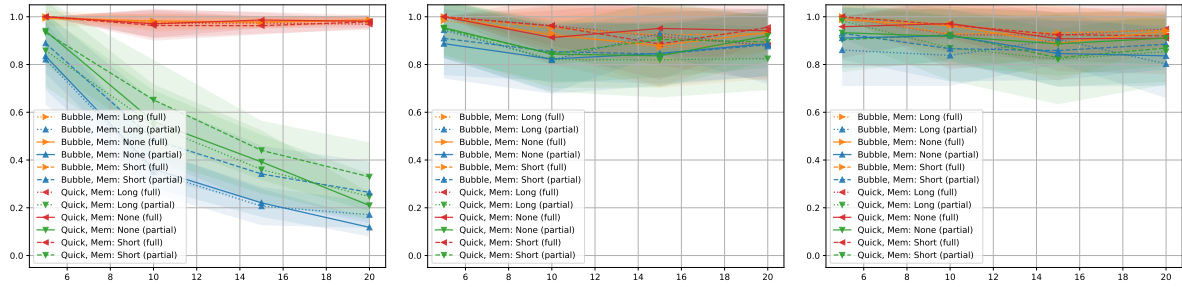


Figure 4: Coaching fidelity against state size for reactive (left), proactive (middle) and reflexive (right) coaching.

the case of reactive coaching, the interplay between fidelity and coverage is clear, with lower advice coverage resulting to the learner eventually reaching the goal state but in a way not conforming to the coach’s search policy, while the more specific the provided advice, the more accurate the learner in its searching steps towards the goal state.

On the contrary, for proactive and reflexive coaching, coaching fidelity is just slightly increasing with respect to advice coverage, being relatively high, even for lower values of advice coverage. Compared to reactive coaching, proactive and reflexive coaching appear to guarantee a constant high fidelity, mitigating the negative effects of low coverage. In a sense, intervening early on guarantees that the learner cannot deviate much from the coach’s guidance, even in case where advice is rather minimal, allowing space for exploration.

5 Discussion and future work

>>> To be completed. . .

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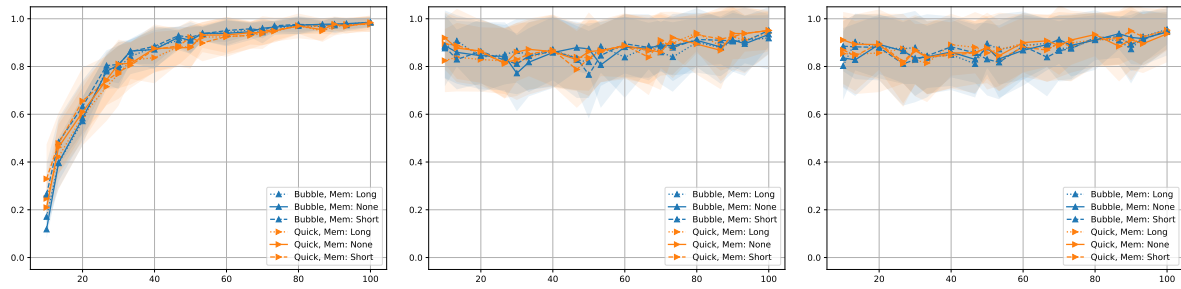


Figure 5: Coaching fidelity against advice coverage for reactive (left), proactive (middle) and reflexive (right) coaching.

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